SENTINEL - Project Architecture Summary

Technology Stack

Frontend

- Next.js 16 (React)
- TypeScript
- Zustand
- Framer Motion
- CSS Variables

Backend

- FastAPI (Python 3.11)
- SQLAlchemy
- Alembic
- SlowAPI
- Python JSON Logger

Database

- PostgreSQL

Caching

- Redis

Deployment

- Docker
- Docker Compose

Development Tools

- VS Code
- Git
- GitHub
- Figma

Core Modules

- Employment Fraud Detector
- Recruiter Scam Detector
- Phishing & URL Analyzer
- Identity / Impersonation Detector
- AI Content Detector
- News Credibility Analyzer
- Explainability Engine

- Dashboard & Scan History

Database Schema

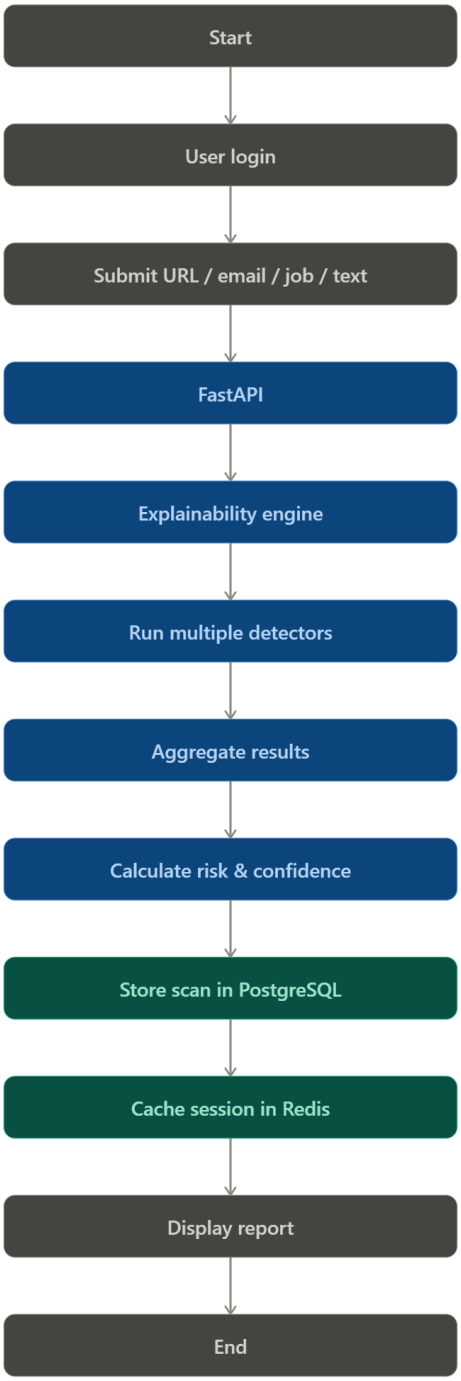
Users
user_id, name, email, password_hash, subscription

Scans
scan_id, user_id, scan_type, risk_score, confidence, json_result

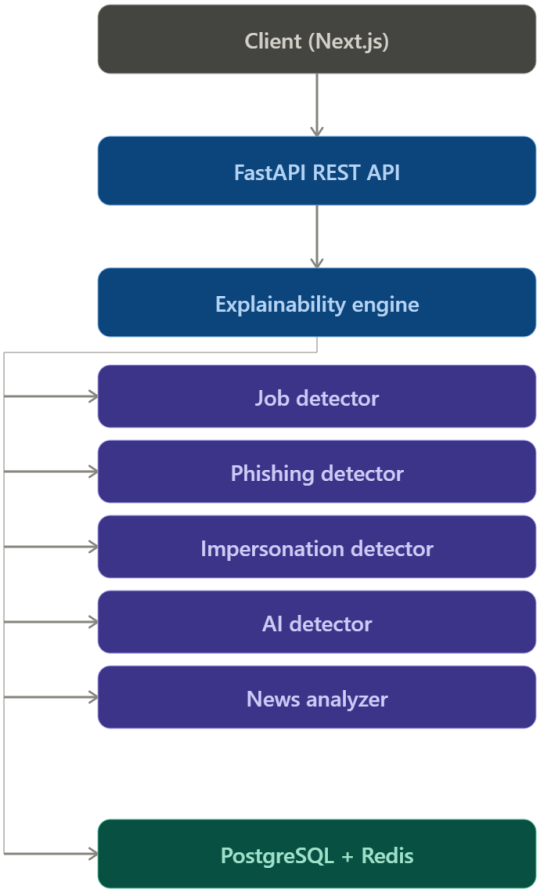
API_Keys
key_id, user_id, api_key_hash, created_at

Sessions (Redis)
session_id, token, expiry

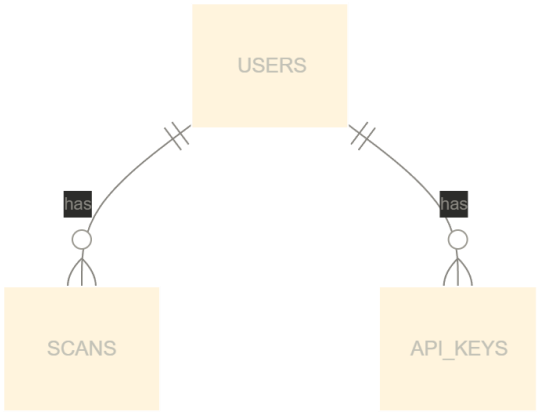
System Flowchart



System Architecture



ER Diagram Overview



Redis stores Sessions and Cache.